

A Five-Point Counterexample to Concavity of Determinantal Entropy

Review note, 5 September 2026

Abstract

Lyons asked whether the Shannon entropy of a finite determinantal probability measure is concave in its positive-contraction kernel. We give a counterexample on five labelled points when complex Hermitian kernels are allowed. The construction comes from a rank-two projection and a purely imaginary direction that is invisible to first order in every event probability. The usually negative Fisher-information term therefore vanishes, while the second-order bending of the probability map has positive entropy pairing. Four short integer vectors specify the example, and a dependency-free exact checker proves $\frac{1}{2} < D^2 H_K[A, A] < \frac{3}{5}$ at the explicit interior kernel.

1 The question and the answer

Let $E = [5]$. If K is a Hermitian matrix with $0 \preceq K \preceq I$, the determinantal probability measure $\mathbb{P}^K$ on subsets of E is defined by

$$\mathbb{P}^K(T \subseteq X) = \det K[T].$$

Writing $p_S(K) = \mathbb{P}^K(X = S)$, its entropy is

$$H(K) = - \sum_{S \subseteq E} p_S(K) \log p_S(K).$$

The conjecture that H is concave in K was posed by Lyons [1] and restated as Conjecture 2.6 in his ICM survey [2]. The formulation there permits real or complex Hilbert spaces.

Here is the counterexample. Put

$$\begin{aligned} \mathbf{1} &= (1, 1, 1, 1, 1)^T, & x &= (-2, -1, 0, 1, 2)^T, \\ q_2 &= (2, -1, -2, -1, 2)^T, & q_3 &= (-1, 2, 0, -2, 1)^T, \end{aligned}$$

and write $u \wedge v = uv^T - vu^T$. Define

$$P = \frac{1}{5} \mathbf{1}\mathbf{1}^T + \frac{1}{10} xx^T, \quad B = \mathbf{1} \wedge x - \frac{1}{2} q_2 \wedge q_3, \quad A = iB, \quad (1)$$

and

$$\varepsilon = 10^{-6}, \quad K = \varepsilon I + (1 - 2\varepsilon)P. \quad (2)$$

Theorem 1. *The matrices in (1)–(2) satisfy $0 < K < I$, $A = A^*$, and*

$$\frac{1}{2} < D^2 H_K[A, A] < \frac{3}{5}.$$

Consequently H is not concave on the complex Hermitian positive contractions on $\mathbb{C}^5$.

The rest of the note explains both the mechanism and the exact verification. The displayed decimal value, included only for orientation, is

$$D^2 H_K[A, A] = 0.56612746277851002589 \dots$$

2 Why a purely imaginary direction is useful

The complete-event probabilities admit the signed determinant formula

$$p_S(K) = (-1)^{|E \setminus S|} \det(K - I_{E \setminus S}), \quad (3)$$

where I_T is the diagonal coordinate projection onto T . This follows directly from Möbius inversion of the inclusion probabilities.

Let $\Phi(K) = (p_S(K))_S$ be the map from kernels to the probability simplex. At an interior kernel, differentiating entropy twice gives

$$D^2 H_K[A, A] = - \sum_S \frac{(Dp_S(K)[A])^2}{p_S(K)} - \sum_S D^2 p_S(K)[A, A] \log p_S(K). \quad (4)$$

The first sum is the pullback of the Fisher metric and is always non-positive. The second sum records the curvature of the probability map in the direction of the entropy gradient.

The construction makes the first sum disappear. If K is real symmetric and B is real skew-symmetric, then

$$\det(K - I_{E \setminus S} + itB) = \det(K - I_{E \setminus S} - itB),$$

because transposition changes t to $-t$. Thus every event probability is an even function of t , and

$$Dp_S(K)[iB] = 0 \quad (S \subseteq E). \quad (5)$$

If

$$b_S(K, B) = \left. \frac{d^2}{dt^2} p_S(K + itB) \right|_{t=0},$$

then (4) reduces to

$$D^2 H_K[iB, iB] = - \sum_S b_S(K, B) \log p_S(K). \quad (6)$$

In information-geometric language, iB lies in the differential kernel of Φ ; the sign is decided entirely by the vertical second-order bending of Φ .

3 How the four vectors were chosen

The vectors $\mathbf{1}, x, q_2, q_3$ are pairwise orthogonal across the two displayed pairs: $\mathbf{1} \perp x$, and $q_2, q_3 \perp \mathbf{1}, x$. Since $\|\mathbf{1}\|^2 = 5$ and $\|x\|^2 = 10$, the matrix P is the orthogonal projection onto $\langle \mathbf{1}, x \rangle$. The two summands of B rotate, respectively, inside $\text{ran}(P)$ and inside $\text{ker}(P)$. Hence

$$P^2 = P = P^T, \quad B^T = -B, \quad PB = BP. \quad (7)$$

It follows at once that K has eigenvalue $1 - \varepsilon$ on $\text{ran}(P)$ and eigenvalue ε on $\text{ker}(P)$. Thus $0 < K < I$, while $A = iB$ is Hermitian.

The two pieces of B are both needed. A rotation confined to only one side of the projection gives a logarithmic divergence in (6). The relative coefficient $-\frac{1}{2}$ balances the two sides so that the divergent terms cancel, leaving a finite positive limit.

4 The positive boundary limit

For $0 < \delta < 1/2$, set $K_\delta = \delta I + (1 - \delta)P$. The coordinates of x are distinct, so every two-row minor of the matrix with columns $\mathbf{1}, x$ is nonzero. Consequently the rank-two projection P is full spark. For each subset S there is a positive rational number a_S such that

$$p_S(K_\delta) = a_S \delta^{w_S} (1 + O(\delta)), \quad w_S = ||S| - 2|. \quad (8)$$

More explicitly,

$$a_S = \det P[S] \quad (|S| \leq 2), \quad a_S = \det(I - P)[E \setminus S] \quad (|S| \geq 2).$$

These formulas follow either from the spectral Bernoulli mixture for K_δ , or directly from principal-minor expansion and Jacobi's complementary-minor identity.

Let $b_S = b_S(P, B)$. Direct second-order determinant expansion gives the two exact cancellations

$$\sum_S b_S = 0, \quad \sum_S w_S b_S = 0. \quad (9)$$

The first is forced by $\sum_S p_S = 1$. The second is the balancing condition between the rotations in $\text{ran}(P)$ and $\text{ker}(P)$. Equations (6)–(9), together with $\delta \log \delta \rightarrow 0$, now give

$$\begin{aligned} \lim_{\delta \downarrow 0} D^2 H_{K_\delta}[iB, iB] &= - \sum_S b_S \log a_S \\ &= \frac{1}{5} \log \frac{2^{260} 3^{143} 5^{350}}{7^{87} 13^{284}} = 0.57665861803438136898 \dots \end{aligned} \quad (10)$$

This is an exact positive number: the numerator inside the logarithm is a 392-digit integer, whereas the denominator is a 390-digit integer.

For transparency, Table 1 groups the 32 determinant terms by their value of a_S . Its second column is $B_a = \sum_{S: a_S=a} b_S$; hence $-\sum_a B_a \log a$ reproduces (10). The ungrouped table is supplied with the checker.

Table 1: Grouped data for the boundary logarithmic sum.

a	B_a	a	B_a
1/50	72/5	2/5	−2
2/25	−204/5	12/25	128/5
3/25	41/5	13/25	284/5
9/50	−606/5	3/5	120
1/5	20	7/10	−32
7/25	247/5	4/5	−36
3/10	60	1	−100
8/25	−112/5		

5 An exact interior certificate

The positive limit already proves that all sufficiently small δ are counterexamples. To remove the hidden threshold, Theorem 1 fixes $\varepsilon = 10^{-6}$. The following verification uses rational arithmetic throughout.

For a positive rational $r \in [1, 2]$, set $y = (r - 1)/(r + 1)$. The positive series for $2 \operatorname{atanh}(y)$ gives, for every positive integer m ,

$$2 \sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1} \leq \log r \leq 2 \sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1} + \frac{2y^{2m+1}}{(2m+1)(1-y^2)}. \quad (11)$$

Every positive rational probability is written as $2^k r$ with $1 \leq r < 2$. Taking $m = 24$ in (11) gives directed rational intervals for all logarithms in (6).

The event probabilities and second derivatives are also rational. They are computed by expanding each five-by-five determinant over its 120 permutations and extracting the coefficients of t^0 and t^2 . Summation checks

$$p_S(K) > 0 \quad \text{for all } S, \quad \sum_S p_S(K) = 1, \quad \sum_S b_S(K, B) = 0.$$

Applying the lower logarithm endpoint to positive coefficients and the upper endpoint to negative coefficients, with the choices reversed for the upper bound, proves

$$0.5661274627785100258893 < D^2 H_K[iB, iB] < 0.5661274627785100258923. \quad (12)$$

The endpoints in the computation are exact fractions; the decimals in (12) are shortened outward displays. In particular, $\frac{1}{2} < D^2 H_K[iB, iB] < \frac{3}{5}$.

Because K is an interior positive contraction, $K + tA$ remains a positive contraction for all sufficiently small real t . A concave twice differentiable function cannot have positive second derivative along such an interior line. This completes the proof of Theorem 1.

The exact checker uses only the Python standard library. It is available with the ungrouped boundary table at

[https://github.com/randomcat4/icm-conjecture-results/tree/release/
I05-lyons-counterexample-20260905/results/I05_lyons_counterexample/author_review_
note.](https://github.com/randomcat4/icm-conjecture-results/tree/release/I05-lyons-counterexample-20260905/results/I05_lyons_counterexample/author_review_note)

6 Scope

The midpoint K is real symmetric, but the direction $A = iB$ is purely imaginary. The example therefore disproves the conjecture with the real-or- complex quantifier used in the cited sources. It does not decide whether entropy is concave when kernels and directions are required to be real symmetric, and it does not establish that five is the smallest possible dimension.

The calculation also suggests a broader structural separation near projection kernels: the Fisher term can enforce negative curvature in all real directions while commuting imaginary directions retain a finite positive boundary limit. A companion note in the repository proves such a separation for an explicit one-parameter family. That extension is not needed for the counterexample above.

We make no absolute priority claim. To the best of our knowledge, the cited conjecture had no previously published proof or counterexample; private communications, seminars, unpublished notes, and unsuccessful attempts are outside the scope of our search.

References

- [1] Russell Lyons. Determinantal probability measures. *Publications Mathématiques de l'IHÉS*, 98:167–212, 2003.
- [2] Russell Lyons. Determinantal probability: Basic properties and conjectures. In *Proceedings of the International Congress of Mathematicians 2014*, volume IV, pages 137–161, 2014.